



BIOLOGICAL OXIDATION

INTRODUCTION TO BIOLOGICAL OXIDATION:

- Oxidation, which occurs in living systems is called biological oxidation.
- Biological oxidations are exergonic.
- During biological oxidations, the reacting chemical systems move from a higher energy level to a lower one and therefore there is liberation of energy.
- The energy released as heat is converted to chemical energy by formation of energy rich compound ATP.
- The formation of ATP from ADP and Pi is termed phosphorylation, as phosphorylation is coupled with biological oxidation, the process is called biological oxidative phosphorylation.

COENZYME SYSTEM INVOLVED IN BIOLOGICAL OXIDATION:

Biological oxidation is brought about with different **enzymes** and **coenzymes**.

- The **enzymes** required for biological oxidation belong to the class of oxidoreductases, which includes:
 - Oxidases
 - Dehydrogenases
 - Hydroperoxidases
 - Oxygenases.
- **Coenzymes** involved in biological oxidation include:
 - Nicotinamide adenine dinucleotide (NAD⁺)
 - Nicotinamide adenine dinucleotide phosphate (NADP⁺)
 - Flavin mononucleotide (FMN)
 - Flavin adenine dinucleotide (FAD).

OXIDASES:

Oxidases catalyze the removal of hydrogen from a substrate in the form of H₂O or H₂O₂ (hydrogen peroxide), using oxygen as a hydrogen acceptor.

- Ex:**
- Cytochrome oxidase
 - L-amino acid oxidases
 - Xanthine oxidase.

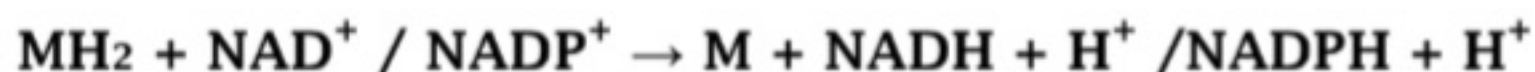
DEHYDROGENASES:

This group constitutes several enzymes. They all catalyze the removal of hydrogen from a substrate but are not able to use oxygen as a hydrogen acceptor. These enzymes, therefore, require specific coenzymes as acceptors of hydrogen atoms. The coenzymes of dehydrogenases may be either:

- Nicotinamide coenzymes (NAD⁺ or NADP⁺)
- Flavin coenzymes (FMN or FAD).

Nicotinamide coenzymes (NAD⁺ or NADP⁺) linked dehydrogenases:

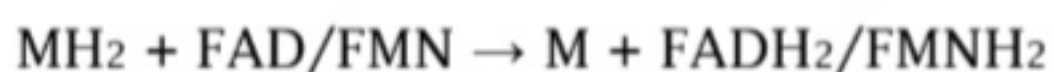
- Some dehydrogenases can use either nicotinamide adenine dinucleotide (NAD⁺) or nicotinamide adenine dinucleotide phosphate (NADP⁺) coenzymes. These coenzymes are derived from vitamin niacin.
- NAD⁺ or NADP⁺ linked dehydrogenases remove two hydrogen atoms from their substrate. One of these is transferred to the NAD⁺ or NADP⁺. The other appears as hydrogen ion (H⁺) in the medium. The general reaction can be written as:



- NAD linked dehydrogenases are involved in the oxidative pathways of metabolism like in glycolysis, TCA cycle and in the mitochondrial respiratory chain.
- NADP linked dehydrogenases, on the other hand, are involved in reductive biosynthetic reactions like fatty acid synthesis and cholesterol synthesis.
- Unlike NADH, NADPH cannot be oxidized with concomitant production of energy.

Flavin-coenzyme (FMN or FAD) linked dehydrogenases:

- Flavin mononucleotide (FMN) and flavin adenine dinucleotide (FAD) are derived from vitamin **riboflavin**.
- Unlike NAD⁺, both hydrogen atoms from substrate are accepted by FMN or FAD. The general reaction can be written as:



- Most of the FMN linked dehydrogenases are concerned with mitochondrial electron transport chain, **Ex:** NADH-dehydrogenase.

- The examples of FAD linked dehydrogenases are succinate dehydrogenase in TCA cycle, acyl-CoA dehydrogenase in β -oxidation of fatty acid, etc.

HYDROPEROXIDASES:

Hydroperoxidases catalyze the reduction of H_2O_2 to H_2O . There are two types of hydroperoxidases:

1. Peroxidases
2. Catalases.

OXYGENASES:

Oxygenases are a group of enzymes that catalyze the addition of one or both of the atoms of the O_2 molecule into the substrate. **Oxygenases** are not concerned with energy production in the body. These are:

MONO-OXYGENASE:

- These enzymes incorporate one oxygen atom of O_2 into the substrates in the form of hydroxyl group, while the other oxygen atom of O_2 is reduced to H_2O .
- An example of monooxygenase is phenylalanine hydroxylase for the formation of tyrosine from phenylalanine.

DIOXYGENASES:

These catalyze the incorporation of both the atoms of O_2 into the substrates. For example, the enzymes such as:

- Homogentisate oxidase of tyrosine metabolism.
- 3-Hydroxy anthranilate oxidase and L-tryptophan dioxygenase of tryptophan metabolism.
- Cyclooxygenase involved in prostaglandin synthesis.



THANK YOU

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